Deliverable 4

Generative Artificial Intelligence (580694)

Due: November 30, 2026, 23:59
Weight: 40% of the semester project

1. Objective

Your team fine-tunes the model behind at least one agent of the Deliverable 3 system, on data built against the failure diagnosed in Deliverable 1, and shows what the new weights achieve that prompting alone did not.

2. Requirements

Choice of agent. Choose the agent with evidence from Deliverable 3: the execution traces should show that this agent is where the errors originate or where prompting stopped helping.

Model constraint. The fine-tuned agent starts from the same open-weight base model. The other agents may keep the base model.

Training data and procedure. Build a dataset of input and output pairs for the role of the fine-tuned agent. State its source, its size, and how each correct output was produced. Data generated by a larger model must name that model, and the team checks a stated sample by hand. The resulting weights or adapter are published, on Hugging Face.

Failure case. Identify at least one input where the fine-tuned system still breaks or degrades, explain why, and store its execution trace in the repository.

3. Format and submission

Video, maximum three minutes. A screen recording of the system with the fine-tuned agent running on real inputs, with the intermediate messages between agents visible. Show the base agent and the fine-tuned agent on the same input. Submit a link with open access.

Technical document. A vertical, single-page PDF compiled from LaTeX. Multi-page submissions and PDFs not produced in LaTeX will be rejected. The page must contain the agent graph with the fine-tuned agent marked, a description of the training data, the training configuration with its loss curves, and the results table for the four systems and the control.

Repository. The repository includes the training script, the training data or a script that rebuilds it, the link to the published weights or adapter, the prompt of each agent as a separate versioned file, and the execution traces.

Hugging Face model. A link with open access to the fine-tuned weights or adapter published on Hugging Face. The weights at that link must be the ones that produced the reported results.

4. Evaluation rubric

Continuity with Deliverables 1 to 3 (10 points). The deliverable attacks the same task and the same failure, and builds on the Deliverable 3 agent system.

Choice of agent (10 points). The fine-tuned agent is chosen with evidence, and the team explains why fine-tuning, and not further prompting, is the intervention this agent needs.

Training data (10 points). The source, size, and labeling of the data are stated, generated outputs are checked by hand on a stated sample, and the separation from the evaluation set is demonstrated.

Training procedure (10 points). The method, hyperparameters, loss curves, training time, and hardware are reported, and the published weights or adapter load and reproduce the reported results.

Evidence of improvement (10 points). Results are reported over a stated set of inputs for the baseline.

Repository and reproducibility (10 points). The link works at the time of grading, and the README allows a reader to reproduce what the video shows.